

Voice-to-3D: Generative AI for Real-Time Generation of 3D Objects for Augmented Reality

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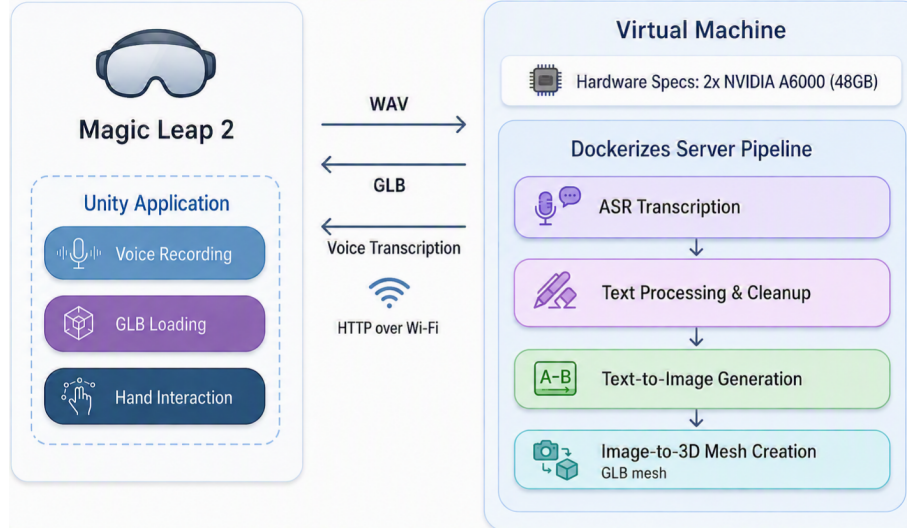


Figure 1: System overview: the user speaks the desired object into the Magic Leap 2 headset, which transmits the audio to a Dockerized GPU server. The pipeline performs automatic speech recognition (ASR), text-to-image generation, and image-to-3D reconstruction, returning a textured GLB mesh rendered as a hologram in approximately three seconds, along with the transcribed text shown as soon as the audio is sent.

ABSTRACT

While augmented reality (AR) increases worker productivity, application development is heavily bottlenecked by the need to pre-create 3D assets. This paper presents an end-to-end pipeline generating 3D assets in real time on an AR application: the user verbally describes an asset, and within three seconds the corresponding GLB mesh appears in their field of view. The backend components were selected via an objective benchmark of four image-to-3D models and a text-to-image preference study ($n = 35$). A System Usability Scale (SUS) evaluation was conducted ($n = 18$) on the final AR application, yielding a score of 82.64, complemented by application-specific feedback.

Index Terms: Mixed reality, 3D generation, voice interface, generative AI, hand interaction, spatial computing.

1 INTRODUCTION

Augmented Reality (AR) is moving into everyday industrial and consumer use, yet the value of an AR application depends on whether it can present the specific digital content a given task requires. The clearest demonstrations of this value are in industry, where AR guidance has been shown to reduce assembly time and

error rates substantially [14, 17, 18]. However, in all cases, the experience is only as rich as the 3D assets available to it.

This is where current applications are constrained. The 3D content an AR experience can display must generally be modelled in advance and bundled with the application, limiting interaction to a fixed, predefined catalogue. Authoring those assets by hand is slow and expensive, creating a content bottleneck that caps how dynamic and open-ended an AR experience can be. Generative AI offers a way out: feed-forward image-to-3D reconstruction networks [4, 25] and fast text-to-image diffusion models [12, 19] can now produce textured 3D objects in seconds rather than hours, albeit at lower geometric and texture fidelity than professionally authored assets. However, what has not yet been demonstrated is their integration into a single pipeline that lets an AR user create arbitrary 3D content on demand, by voice, at real-time speed [24, 3, 2, 27], that is, an interactive rate fast enough to sustain a continuous speak-and-see sequence, keeping the user’s attention [16, 7].

We present a voice-to-3D pipeline that closes this gap through a client-server architecture, offloading inference to a remote GPU, since these models remain too heavy for standalone AR hardware. The user opens a hand-anchored menu and describes an object by voice (for example, “a wooden chair”); a remote GPU server performs automatic speech recognition (ASR), text-to-image generation, and image-to-3D reconstruction; and the resulting textured GLB mesh is streamed back and rendered as a hologram that can be grabbed, moved, rotated, and scaled with bare-hand gestures. The end-to-end latency is approximately three seconds, and every model in the chain was chosen through dedicated evaluation rather than assumption. The central challenge is therefore not building a faster component, but a principled way to assemble the existing

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Figure 2: Voice prompts mapped to 3D holograms generated dynamically in the Magic Leap 2 workspace.

ones into a chain that stays within an interactive latency budget, with the evidence that the result is usable in practice. Fig. 2 shows some of the objects created with our application.

Our contributions are: (1) a complete pipeline where the voice of the user turns into an interactible 3D object in an AR headset; (2) a systematic benchmark of four image-to-3D models characterizing the quality–speed trade-off for interactive AR, with a speed-constrained criterion for selecting among them, including an export optimization that cuts TRELIS generation from ~ 44 s to ~ 10 s; (3) a blind text-to-image preference study ($n = 35$) validating the image-generation stage; and (4) a usability evaluation of the final application ($N = 18$) measuring the usability and experience of the complete system using the System Usability Scale and application-specific metrics.

2 RELATED WORK

Generative 3D content. Early text-to-3D methods such as DreamFusion [20] optimize a neural radiance field [15] against a 2D diffusion prior, producing high-quality objects but requiring minutes to hours per asset, far outside any interactive budget. Direct text-to-3D models such as Shap-E [10] generate objects in seconds, but their outputs are low-resolution and untextured. The current real-time direction is instead a *two-stage* one: a fast text-to-image model produces a single image, which a feed-forward network then lifts into a textured mesh. Feed-forward reconstructors such as TripoSR [23], SF3D [4], TRELIS [25] and TRELIS 2 [26] achieve this in sub-second to multi-second times. A systematic evaluation of speech-driven 3D pipelines for AR found that image-intermediate pipelines outperform direct text-to-3D methods on complex or uncommon prompts, while direct methods remain competitive on common objects [27]. A likely reason is the relative scarcity of 3D training data, which remains expensive to acquire at scale [8].

Fast text-to-image. The image stage has likewise accelerated. Latent diffusion models [21] and SDXL [19] established high-quality open generation, and distilled variants such as SDXL-Lightning [12] cut inference to a few steps. More recent models such as FLUX.2[Klein] [11] and Z-Image-Turbo [6, 13, 9] push this further, pairing strong prompt alignment with sub-second generation, making them well suited to an interactive pipeline.

3D generation in AR/XR. Several systems have paired generative models with immersive interfaces. Dream Mesh [24] couples Whisper with DreamFusion for speech-to-3D in mixed reality, but its optimization-based backbone needs tens of minutes per object and it reports no user study. More recent work uses faster models: “Transcending Dimensions” [3] reconstructs objects with Shap-E (a reported ~ 68 s end-to-end), and the Matrix framework [2] generates from speech with an LLM and Shap-E in under 50 s. The most directly comparable effort, “Say It, See It” [27], systematically evaluates four speech-to-3D pipelines on a Quest 3 and confirms that image-intermediate (two-stage) pipelines outperform direct text-to-3D on complex prompts, but its fastest pipeline still takes ~ 18 s and its best-quality one up to ~ 74 s. Across these,

Table 1: Speech-driven 3D generation systems for AR/MR. Our system attains the lowest end-to-end latency while providing a validated usability score.

System	Input	Latency	Hardware	Study
Dream Mesh [24]	Speech	~ 30 – 40 m	RTX 3090 Ti	None
Transc. Dim. [3]	Image	~ 68 s [†]	N/S	SUS 69.6
Matrix [2]	Speech	< 50 s	Tesla T4	SUS 69.6
Say It See It [27]	Speech	18–74 s	3×RTX 3090	Likert
Ours	Speech	~ 3 s	2×A6000	SUS 82.6

[†] Sum of reported subsystem times. N/S: not specified.

a consistent gap remains: none combines few-second, end-to-end voice-to-3D generation with full hand-tracked manipulation and a validated usability evaluation on a see-through headset. Tab. 1 summarizes the comparison.

3 MODEL SELECTION

Each stage was selected by measurement rather than assumption. Every candidate was containerized behind a uniform HTTP interface, letting pipeline combinations be swapped and tested through a browser-based orchestration tool. The text-to-image and image-to-3D stages are evaluated below; the ASR stage used Parakeet TDT, chosen from published benchmarks as the fastest open source model available.

3.1 Text-to-Image Preference Study

Three candidate models were first compared visually at the 512×512 resolution required by the image-to-3D stage. SDXL-Lightning was excluded at this point: trained at 1024×1024 , it produces tiling and duplication artifacts when forced to 512×512 , leaving FLUX.2[Klein] and Z-Image-Turbo as the viable candidates. These two were then compared in a blind study with 35 participants, who chose their preferred image across five prompts without knowing which model produced each (175 comparisons total). Z-Image-Turbo was preferred in 54.9% of cases versus 41.1% for FLUX.2[Klein] (4.0% undecided). Combined with its faster inference (~ 1.5 s vs. ~ 2 s), Z-Image-Turbo was selected for the pipeline.

3.2 Image-to-3D Benchmark

We benchmarked four feed-forward reconstructors (TripoSR, SF3D, TRELIS, TRELIS 2), selected because they are the models capable of sub-second to few-second reconstruction and thus the only candidates that could plausibly meet an interactive budget; slower optimization-based methods were excluded by design. The three reference objects were chosen to span increasing geometric complexity, a rounded solid (apple), a thin directional form (arrow), and a multi-part structure (airplane), mirroring the increasing-complexity design used in concurrent AR evaluations [27]. Reconstructions were aligned to the ground-truth mesh by PCA pre-alignment with ICP refinement, and geometric (Chamfer distance, F-score), perceptual (PSNR, LPIPS), and timing metrics were computed from 16 viewpoints on a Fibonacci sphere. Tab. 2 reports the averages, and Fig. 3 shows the corresponding reconstructions for visual comparison. We stress that our goal is not another isolated evaluation of these models: each has already been benchmarked individually in prior work. What is missing is a *controlled, head-to-head* comparison across the same input images, and the same hardware across all models, which lets their reconstructions and elapsed time be compared directly against one another and against a shared ground truth (Fig. 3). This benchmark is therefore not a universal ranking, but deliberately constrained by our hardware and by a speed budget that favours the fastest models capable of interactive generation.

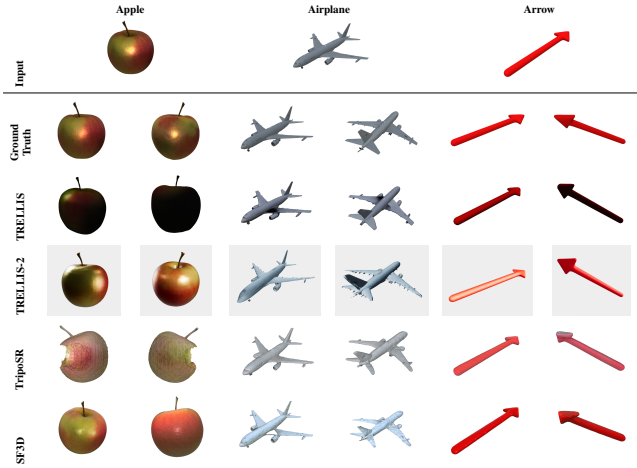


Figure 3: Qualitative image-to-3D reconstructions (two views per object).

Table 2: Average image-to-3D benchmark across the three objects. Best per column in **bold**.

Model	CD-L2 ↓	F@0.02 ↑	PSNR ↑	LPIPS ↓	Time ↓
TriposSR	0.00244	0.648	18.06	0.199	0.46 s
TRELIS	0.00063	0.814	16.09	0.206	7.70 s
SF3D	0.00081	0.827	17.93	0.163	0.50 s
TRELIS 2	0.00043	0.903	20.22	0.146	70.48 s

The benchmark exposes a sharp quality–speed split. TRELIS 2 attains the best score on every quality metric, and its superior fidelity is visible in Fig. 3, but its 70.48 s generation time is two orders of magnitude above an interactive budget. The original TRELIS was profiled in an attempt to close this gap: its GLB export dominated runtime, and switching to fast texture baking and a faster mesh-decimation library cut its total generation from roughly 44 s to 10 s, still well outside the sub-second interactive regime. Of the two sub-second models, SF3D clearly outperforms TriposSR in precision (F@0.02: 0.827 vs. 0.648) and perceptual quality (LPIPS: 0.163 vs. 0.199) at essentially the same speed, and it also surpasses the much slower TRELIS on most quality metrics while being roughly 15× faster. SF3D therefore offers the best reconstruction quality attainable within a real-time budget and was selected for the pipeline.

4 SYSTEM

The system has two parts: a server-side pipeline on a remote GPU workstation, and a client-side AR application on the Magic Leap 2. The teaser (Fig. 1) shows the overall data flow.

4.1 Server Pipeline

The pipeline is deployed as four Docker containers orchestrated with Docker Compose on a workstation with two NVIDIA A6000 GPUs (48 GB each). Three containers host the selected models, each behind an HTTP endpoint: Parakeet TDT for speech recognition (GPU 0), Z-Image-Turbo for text-to-image (GPU 1), and SF3D for image-to-3D reconstruction (GPU 1). A fourth, CPU-only orchestration container receives the audio from the headset, cleans the transcript with a regex pass that strips filler words (e.g. “I want to see a blue apple” → “blue apple”), and forwards it through the remaining two models in sequence. To reduce *perceived* latency, the server replies in two phases: Phase 1 returns the transcript almost immediately (~0.5 s), while Phase 2 generates the image and mesh in the background (~2.5 s). Seeing the recognized words within

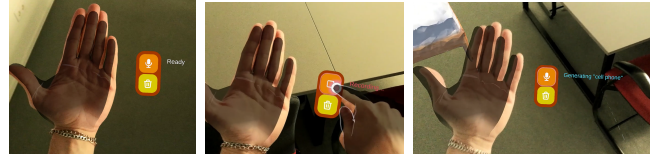


Figure 4: The palm-anchored hand menu seen through the Magic Leap 2. Left to right: ready, recording, and generating states.

half a second fills the wait with feedback instead of a blank pause, making the generation feel shorter than it is.

4.2 AR Client

The client was built in Unity 2022.3 LTS with the Mixed Reality Toolkit 3 (MRTK3), using OpenXR for input abstraction and the Magic Leap SDK for hand tracking. The interface is a palm-anchored hand menu, implemented with MRTK3’s HandConstraintPalmUp solver, that appears when the user opens their palm toward the headset and hides when the hand closes. It exposes a Record/Stop toggle and a Clear button, with a colour-coded status line for feedback (Fig. 4).

The audio is recorded and sent to the server; once Phase 1 is returned, the recognized transcript is shown to the user both as floating text and in the status line, before the model appears. The returned GLB meshes are loaded at run time with glTFast, placed one metre in front of the user, and normalized to a fixed size. Each object is made interactive through MRTK3’s ObjectManipulator: a one-hand pinch translates and rotates it, a two-hand pinch scales it, and objects out of arm’s reach can be grabbed at a distance by pinching the pointing ray (Fig. 5).

5 USER EVALUATION

The complete system was evaluated in a controlled usability study with 18 participants, conducted on the Magic Leap 2. Each participant completed an individual session comprising a short tutorial, a set of guided tasks (generating simple and complex objects by voice, then manipulating and clearing them), and a period of free exploration. The cohort was largely composed of AR novices, providing a strong test of how self-explanatory the interface is to first-time users.

After the session, participants completed a questionnaire combining the standard ten-item System Usability Scale (SUS) [5] with six application-specific Likert items and four open-ended questions for additional feedback. The system obtained a mean SUS score of **82.64**, which corresponds to a grade of A and falls in the upper “Good”-to-“Excellent” band, well above the commonly cited acceptability threshold of ~70 [1, 22]. The application-specific ratings reinforced this result (Tab. 3; overall median 5 of 5).

The palm-anchored menu and the generation wait time were the highest-rated features, the latter confirming that the choice of a sub-second 3D reconstruction model like SF3D matched user expectations in practice. The lowest-rated item was transcription accuracy,

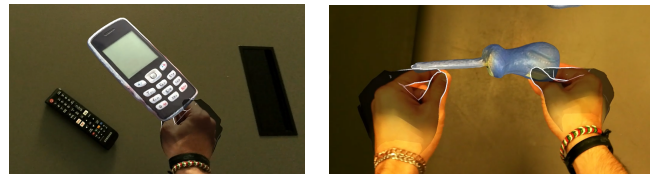


Figure 5: Bare-hand manipulation of a generated object. Left: a one-hand pinch grabs and moves an object. Right: a two-hand pinch resizes it by changing the distance between the hands.

Table 3: Application-specific ratings. Median, IQR, and mean.

Feature	Mdn	IQR	Mean
Hand-palm menu access	5	0.0	4.67
Voice recording	4.5	1.0	4.39
Generation wait time	5	0.0	4.61
3D asset quality	5	1.0	4.39
Object manipulation	4	1.8	4.06
Voice transcription	4	0.8	3.89
Overall	5	1.0	4.33

as the English-only speech model occasionally mis-transcribed the accented input of the native Portuguese-speaking participants, motivating future work. Remaining difficulties were confined to a brief habituation to bare-hand gestures, resolving within the first minutes. The open-ended responses most often praised the creative freedom of generating an arbitrary object from a spoken description rather than choosing from a fixed library (Fig. 6).

6 CONCLUSION AND FUTURE WORK

This paper presented an end-to-end, voice-driven pipeline that generates manipulable 3D assets in real time within an AR application on the Magic Leap 2. A spoken description is transcribed, synthesised into an image, and reconstructed into a textured mesh that appears as a hologram in roughly three seconds, where it can be grabbed, rotated, and scaled with bare hands. Such on-demand generation suits various possible domains, including industrial maintenance and training, rapid prototyping, and interior design. Each stage was chosen through dedicated evaluation: a blind preference study selected the text-to-image model, and a speed-constrained benchmark identified SF3D as the reconstruction model offering the best quality attainable within an interactive budget. A usability study ($n = 18$) returned a mean SUS score of 82.64 (grade A), confirming that the system is usable by first-time users and that real-time, voice-driven 3D generation in AR is feasible today.

These findings, together with the constraints under which the system was built, motivate future work. The reported latency is highly dependent on the GPU server; on more capable hardware, the same speed-constrained selection would admit higher-fidelity models, raising quality at no cost to responsiveness. The evaluation itself could also be broadened: the image-to-3D benchmark used three objects and the preference study five prompts, so a larger study spanning more object categories and prompts, with failure cases and statistical analysis, would strengthen the model-selection evidence. The most requested improvement is native-language ASR, so non-English speakers are not penalised by accent-related mis-transcription, alongside a user-selectable quality-speed mode. Further directions include prompt conditioning for more consistent geometry, surfacing the intermediate 2D image for transparency, and finer per-object scene management.



Figure 6: Free-exploration prompts: “a dinosaur with butterfly wings and purple feet” (left) and “a pink elephant riding on a skateboard” (right).

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